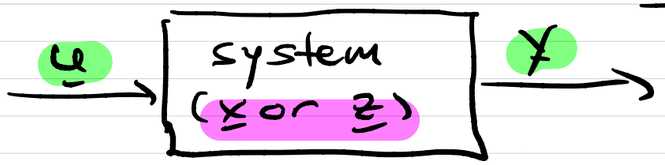
$\dot{\underline{x}} = A\underline{x} + B\underline{u}$ $\quad$ $\dot{\underline{z}} = A_p\underline{z} + B_p\underline{u}$

$\underline{y} = C\underline{x} + D\underline{u}$ $\quad$ $\underline{y} = C_p\underline{z} + D_p\underline{u}$

$A_p = P^{-1}(AP - \dot{P})$ $\quad B_p = P^{-1}B$ $\quad C_p = CP$ $\quad D_p = D$



$\underline{u}$, $\underline{y}$ same in both models

$\Rightarrow$ I/O behavior of system same regardless of coordinates used for states.

Claim: Let $\Psi_x(t)$ be any fund'l matrix for $\underline{x}(t)$ eq'ns

and similarly for $\Psi_z(t)$.

Then $\Psi_x(t) = P(t)\Psi_z(t)$

Pf: $\Psi_x(t) = [\underline{\Psi}_{x,1}(t) \cdots \underline{\Psi}_{x,n}(t)]$

where each $\underline{\Psi}_{x,k}(t)$ satisfies the homog. dynamics for $\underline{x}(t)$

But any sol'n for $\underline{x}(t)$ can be written as $\underline{x}(t) = P(t)\underline{z}(t)$

with $P(t)$ nonsingular

so $\underline{\Psi}_{x,k}(t) = P(t)\underline{\Psi}_{z,k}(t)$ for some $\underline{\Psi}_{z,k}(t)$

$$\Psi_x = [(P\underline{\Psi}_{z,1})\ (P\underline{\Psi}_{z,2})\ \cdots\ (P\underline{\Psi}_{z,n})]$$

$$= P\Psi_z$$

$$\Psi_z = [\underline{\Psi}_{z,1}\ \underline{\Psi}_{z,2}\ \cdots\ \underline{\Psi}_{z,n}]$$

$=$ Claim:

$$\Phi_x(t,T) = P(t)\Phi_z(t,T)P^{-1}(T)$$

Pf/

$$\Phi_x(t,T) = \Psi_x(t)\Psi_x(T)^{-1}$$

$$= [P(t)\Psi_z(t)][P(T)\Psi_z(T)]^{-1}$$

$$= P(t)\Psi_z(t)\Psi_z(T)^{-1}P(T)^{-1}$$

$$= P(t)\Phi_z(t,T)P^{-1}(T)$$

Claim: Impulse response matrix same.

$$G_x(t,\tau) = G_2(t,\tau)$$

Pf:

$$G_x(t,\tau) = C(t)\phi_x(t,\tau)B(\tau) + D(t)\delta(t-\tau)$$

$$= \underbrace{C(t)\left[P(t)\phi_2(t,\tau)P^{-1}(\tau)\right]B(\tau)}_{C_p(t)\qquad\qquad B_p(\tau)} + D(t)\delta(t-\tau)$$

$$= C_p(t)\phi_2(t,\tau)B_p(\tau) + \underbrace{D_p(t)}_{=D(t)}\delta(t-\tau)$$

$$= G_2(t,\tau)$$

= for LTI, this implies the TFM is invariant

$$G_x(s) = G_2(s)$$

$$G_x(s) = C\left[sI - A\right]^{-1}B + D$$

$$A_p = P^{-1}AP \quad (LTI)$$

$$\Rightarrow A = PA_pP^{-1}$$

$$G_x(s) = C\left[sI - PA_pP^{-1}\right]^{-1}B + D$$

Note $I = PP^{-1}$

$$G_x(s) = C\left[P(sI - A_p)P^{-1}\right]^{-1}B + D$$

$$= C\left[P(sI - A_p)^{-1}P^{-1}\right]B + D$$

$$= C_p(sI - A_p)^{-1}B_p + D_p$$

$$= G_2(s)$$

Other useful coord xforms for LTI:

Results in

$$\dot{z}_k = \lambda_k z_k + \sum_{j=1}^{m}(w_k^T b_j)u_j$$

(Decoupled)

Might want to modify if λ_k complex since then sol'ns $z_k(t)$ are also complex and hard to interpret physically.

Example:

Suppose $\lambda_{1,2} = \sigma_1 \pm j\omega_1$

$\lambda_{3,4} = \sigma_3 \pm j\omega_3$

$\lambda_5, \ldots, \lambda_n$ real, non-repeated

There exists nonsingular P such that

$$A_p = \begin{bmatrix} \sigma_1 & \omega_1 & & & & \\ -\omega_1 & \sigma_1 & & & & \\ & & \sigma_3 & \omega_3 & & \\ & & -\omega_3 & \sigma_3 & & \\ & & & & \lambda_5 & \\ & & & & & \ddots \\ & & & & & & \lambda_n \end{bmatrix}$$

"Block modal form"

$$P = \begin{bmatrix} \mathrm{Re}\{\underline{v}_1\} & \mathrm{Im}\{\underline{v}_1\} & \mathrm{Re}\{\underline{v}_3\} & \mathrm{Im}\{\underline{v}_3\} & \underline{v}_5 & \cdots & \underline{v}_n \end{bmatrix}$$

$\lambda_1 = \sigma_1 + j\omega_1 \Rightarrow \underline{v}_1$

$\lambda_2 = \sigma_2 - j\omega_2 \Rightarrow \underline{v}_2$

$\lambda_2 = \bar{\lambda}_1 \Rightarrow \underline{v}_2 = \bar{\underline{v}}_1$

Pure Diagonalization would give

$$\Rightarrow \begin{bmatrix} \sigma_1 + j\omega_1 & & & & & \\ & \sigma_1 - j\omega_1 & & & & \\ & & \sigma_3 + j\omega_3 & & & \\ & & & \sigma_3 - j\omega_3 & & \\ & & & & \lambda_5 & \\ & & & & & \ddots \\ & & & & & & \lambda_n \end{bmatrix}$$

$$e^{A_p t} = \begin{bmatrix} \begin{bmatrix} e^{\sigma_1 t}\cos\omega_1 t & e^{\sigma_1 t}\sin\omega_1 t \\ -e^{\sigma_1 t}\sin\omega_1 t & e^{\sigma_1 t}\cos\omega_1 t \end{bmatrix} & \emptyset & \\ \emptyset & \begin{bmatrix} e^{\sigma_3 t}\cos\omega_3 t & e^{\sigma_3 t}\sin\omega_3 t \\ -e^{\sigma_3 t}\sin\omega_3 t & e^{\sigma_3 t}\cos\omega_3 t \end{bmatrix} & \\ & & \begin{matrix} e^{\lambda_5 t} & & \\ & \ddots & \\ & & e^{\lambda_n t} \end{matrix} \end{bmatrix}$$

and $\underline{x}(t) = P\underline{z}(t) = Pe^{A_p t}\underline{z}_0$

$= z_1(t)\,\mathrm{Re}\{\underline{v}_1\} + z_2(t)\,\mathrm{Im}\{\underline{v}_1\} + z_3(t)\,\mathrm{Re}\{\underline{v}_3\} + z_4(t)\,\mathrm{Im}\{\underline{v}_3\} + z_5(t)\,\underline{v}_5 + \cdots$

oscillations for 1st complex mode occur in a plane spanned by $\{\mathrm{Re}\{\underline{v}_1\}, \mathrm{Im}\{\underline{v}_1\}\}$

2 osc. for 2nd complex mode occur in plane spanned by $\mathrm{Re}\{\underline{v}_3\}, \mathrm{Im}\{\underline{v}_3\}$

Suppose $\underline{z}_0 = \underline{z}(0) = \begin{bmatrix} a \\ b \\ 0 \\ \vdots \\ 0 \end{bmatrix}$

Then $\underline{z}(t) = e^{A_p t}\underline{z}_0$

$z_1(t)$ $\qquad$ $z_2(t)$

$$= a\begin{bmatrix} e^{\sigma_1 t}\cos\omega_1 t \\ -e^{\sigma_1 t}\sin\omega_1 t \\ 0 \\ \vdots \\ 0 \end{bmatrix} + b\begin{bmatrix} e^{\sigma_1 t}\sin\omega_1 t \\ e^{\sigma_1 t}\cos\omega_1 t \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$\Rightarrow$

$$\underline{x}(t) = e^{\sigma_1 t}(a\cos\omega_1 t + b\sin\omega_1 t)\,\mathrm{Re}\{\underline{v}_1\} + e^{\sigma_1 t}(b\cos\omega_1 t - a\sin\omega_1 t)\,\mathrm{Im}\{\underline{v}_1\}$$

$\Rightarrow$ only 1st mode is present, and creates changes in $\underline{x}(t)$ in only 2 directions: $\mathrm{Re}\{\underline{v}_1\}, \mathrm{Im}\{\underline{v}_1\}$

Note that

$$\underline{x}_0 = P\underline{z}_0 = a\,\mathrm{Re}\{\underline{v}_1\} + b\,\mathrm{Im}\{\underline{v}_1\}$$

Thus, if $\underline{x}_0 \in \mathrm{span}\{\mathrm{Re}\{\underline{v}_1\}, \mathrm{Im}\{\underline{v}_1\}\}$

Then

1.) Sol'n contains only 1st mode (oscillations w/ freq ω_1)

2.) $\underline{x}(t) \in \mathrm{span}\{\mathrm{Re}\{\underline{v}_1\}, \mathrm{Im}\{\underline{v}_1\}\}$ for all $t \geq 0$

i.e. the modal oscillations occur in a planar subset of the original state space.

= Similarly $\underline{x}_0 \in \mathrm{span}\{\mathrm{Re}\{\underline{v}_3\}, \mathrm{Im}\{\underline{v}_3\}\}$ would uniquely excite 2nd oscillatory mode, and resulting $\underline{x}(t)$ would be confined to this planar subset of state space.

An alternate visualization:

Let complex eigenvector $\underline{v}$ be expressed in polar form:

$$\underline{v} = \begin{bmatrix} r_1 e^{j\varphi_1} \\ r_2 e^{j\varphi_2} \\ \vdots \\ r_n e^{j\varphi_n} \end{bmatrix}$$

for a mode with such an eigenvector, the corresponding amplitude of oscillations in each component of $\underline{x}(t)$ will have relative amplitude r_i and relative phase φ_i

i.e. ratio of ampl. of oscillations in $x_1(t)$ compared to ampl. of oscillations in $x_3(t)$ would be

r_1/r_3

and diff in phase between osc. in $x_1(t)$ and $x_3(t)$ would be

$\varphi_1 - \varphi_3$

Another reason to find alternate P for LTI:

repeated (real) eigenvalues